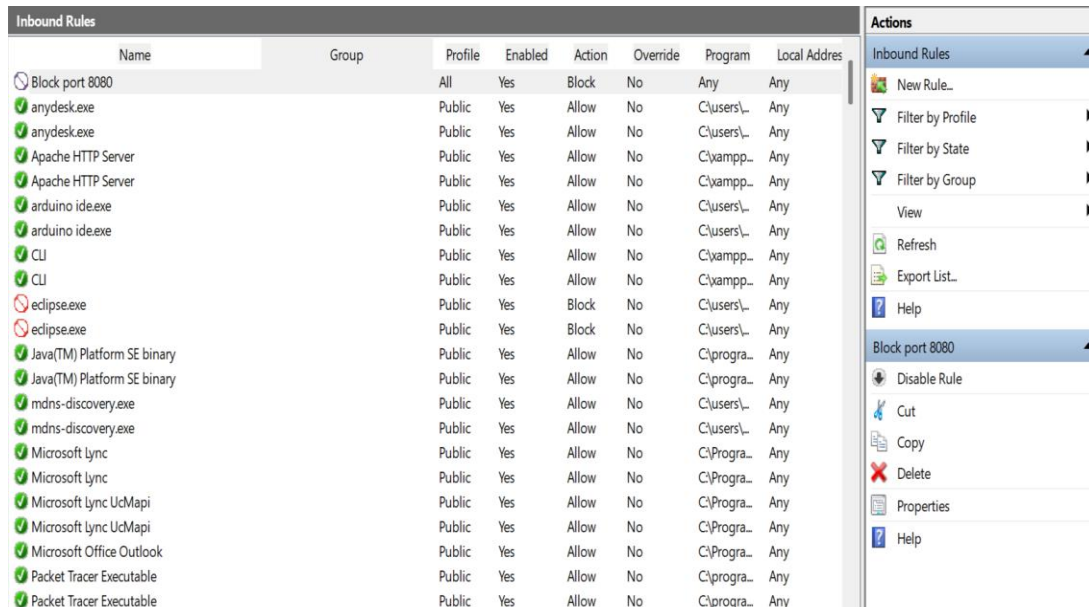
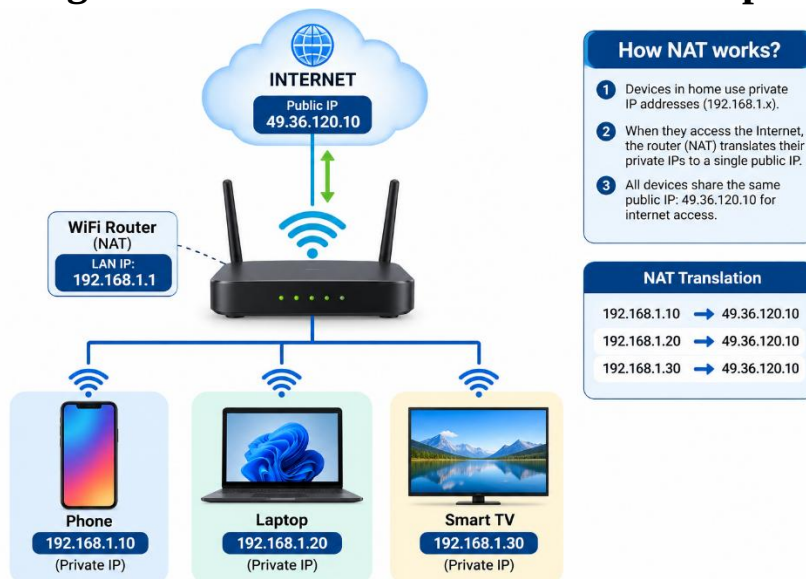


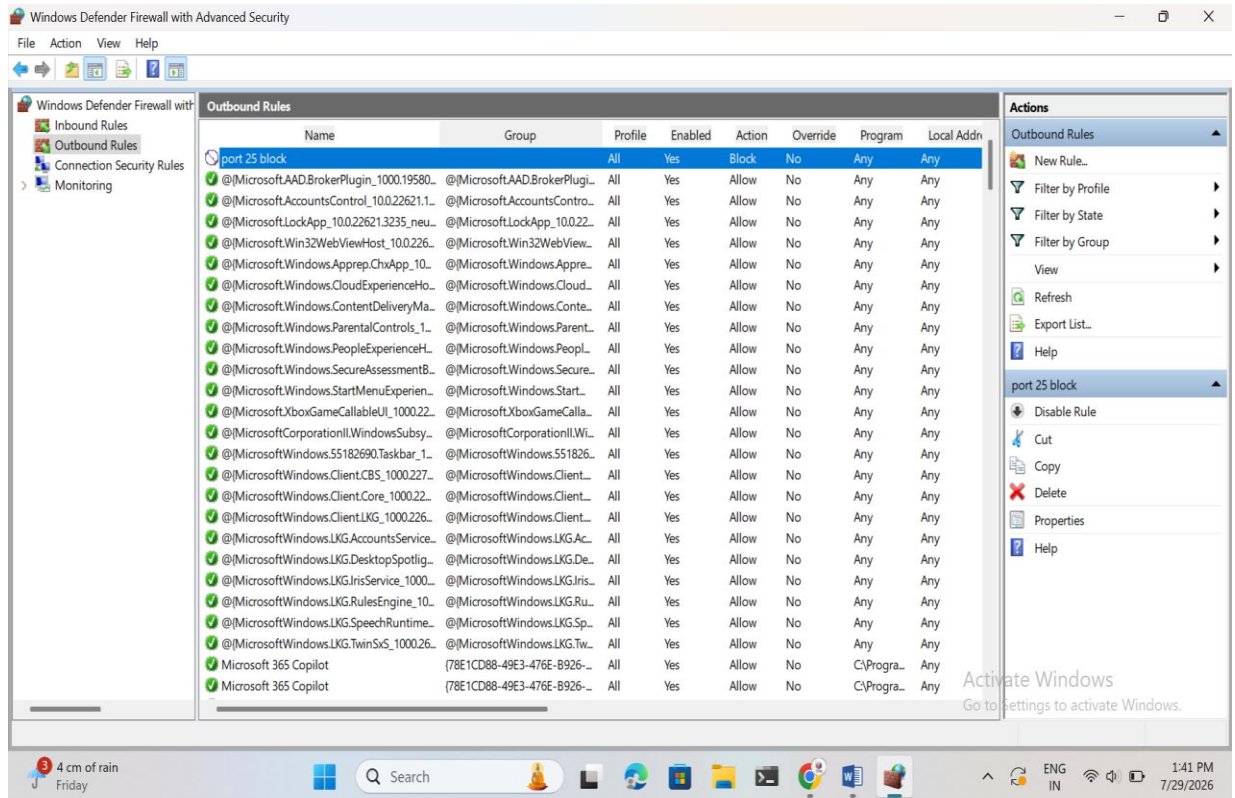
1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access
Hint: Use any drawing tool or even hand-draw and click a photo



3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.



4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53
Hint: Focus on source/destination IPs and ports in the rule set.

No	Packet	Direction/Port	Result	Reason
1	HTTP: 192.168.1.5 TO 172.217.166.78:80	HTTP/Port 80	Blocked	Port 80 is not explicitly allowed as default rule is block.
2	SSH request from 10.0.0.10 to 34.201.23.1:22	Outbound/Port22	Allowed	Rule explicitly says SSH outbound allow.
3	DNS query from 192.168.1.7 to 8.8.8.8:53	Outbound/Port53	Blocked	The rule only allows inbound DNS if it's secure. Outbound DNS is not listed so it remains default block.

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

1. Block Unauthorized Access

Firewalls allow only approved network traffic and block unauthorized connection attempts to protect servers from hackers.

2. Prevent DDoS Attacks

Packet filtering detects and blocks suspicious or excessive traffic, helping protect servers from Distributed Denial-of-Service (DDoS) attacks.

3. Filter Malicious Requests

Firewalls inspect incoming packets and block requests containing malicious payloads, reducing the risk of attacks such as SQL injection, malware, or unauthorized access.